



MANIPAL INSTITUTE OF TECHNOLOGY

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**MINI PROJECT REPORT
PLC AND SENSORICS LAB**

PLC Ladder Program for Automatic Car Wash

Submitted by

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ABSTRACT

This paper explores the development of an Automated Carwash system utilizing a Programmable Logic Controller (PLC). The primary objective is to reduce the volume of water used in car washes and mitigate potential hazards associated with prolonged chemical exposure or possible injuries. In Kuwait, car washing is a significant consumer of desalinated water, with each car wash requiring between 50 and 100 gallons of fresh water. Research indicates that 75% of this water can be filtered and reused, necessitating only about 25% of freshwater for the final rinse and makeup. Given the current global economic challenges, implementing an automated carwash process can minimize the need for human supervision and interaction. In this study, a PLC was programmed and deployed to achieve an efficient automated carwash system.

Keywords: Programmable Logic Controller (PLC), Automated Carwash

INDIVIDUAL CONTRIBUTIONS

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INTRODUCTION

The paper discusses the development of an Automated Car Wash system using a Programmable Logic Controller (PLC). This project is an essential aspect of electrical engineering as it involves the design, creation, and maintenance of electrical control systems, machinery, and equipment. Originally developed by General Motors in 1968 to avoid redesigning control devices with each new car model (Al-Odwani et al., 2007), PLCs have since become integral to various automated systems.

Car technology, a vital means of transportation, communication, and business, has led to an increase in the number of cars in urban areas globally (Ridley et al., 2003). This surge has necessitated the development of automated car wash systems utilizing PLC controllers. The primary aim of this project is to reduce water usage in car washes and mitigate potential hazards from prolonged chemical exposure or injuries. In places like Kuwait, car washes consume significant amounts of desalinated water, with each wash requiring between 50 and 100 gallons of fresh water.

Research indicates that 75% of this water can be filtered and reused, reducing the need for freshwater to just 25% for the final rinse. This automated system, by reusing water, can achieve substantial water savings and promote environmental sustainability (Bala et al., 2020).

The research focuses on the operation of PLCs as controllers, the programming languages used by PLCs to manage systems, and the implementation of an automated car washing machine. This designed automatic car wash system saves time and money, reduces water consumption, and minimizes manpower, thus preventing exposure to hazardous chemical substances in detergents (May et al., 2019).

The system's efficiency lies in its structured process: each phase, from detecting the car at the entrance to the final drying section, takes only one minute. Water consumption is minimized by spraying only when sensors detect a car, unlike manual washing, which often results in water wastage. This project can save up to 75% of the water used, reusing it in the rinse section. Additionally, the automated process reduces the need for human supervision and interaction, addressing current global economic challenges.

LITERATURE SURVEY

Previous studies highlight the extensive water consumption in car washing and the potential for water reuse through filtration. The literature indicates a growing trend towards automation in various industries to enhance efficiency and reduce labor costs. The integration of PLCs in automated systems has shown significant improvements in process control and reliability. This project builds on these findings by applying PLC technology to automate the carwash process, aiming to achieve both water conservation and operational efficiency.

METHODOLOGY

This section discusses the proposed Automated Car Wash with a PLC Controller, encompassing software development and hardware construction. The software development focuses on system operation and simulation, while the hardware construction involves building a prototype.

To achieve optimal results in circuit design, adhering to the PLC's specifications and programming is crucial. This ensures flawless performance and prevents failures. Accurate coding is essential; errors can cause the prototype to malfunction. Extra attention must be paid to these critical aspects.

illustrates the operation of the Automated Car Wash system using a PLC Controller. The process begins when the Switch Button is triggered or Sensor 1 detects a car, activating the conveyor and water spray after a 20-second delay. Excess water is filtered and reused in the rinse section, while the green light turns off and the red light turns on.

A) RESOURCES UTILIZED (SOFTWARE / HARDWARE DETAILS)

- **PLC Model:** BOSCH REXROTH INDRAL20
- **Software:** IndraWorks
- **Sensors:** Infrared car detection sensors
- **Actuators:** Conveyor motors, washer/brush motors, rinse valves, dryer fans
- **Miscellaneous:** Memory bits, control lights (red and green)

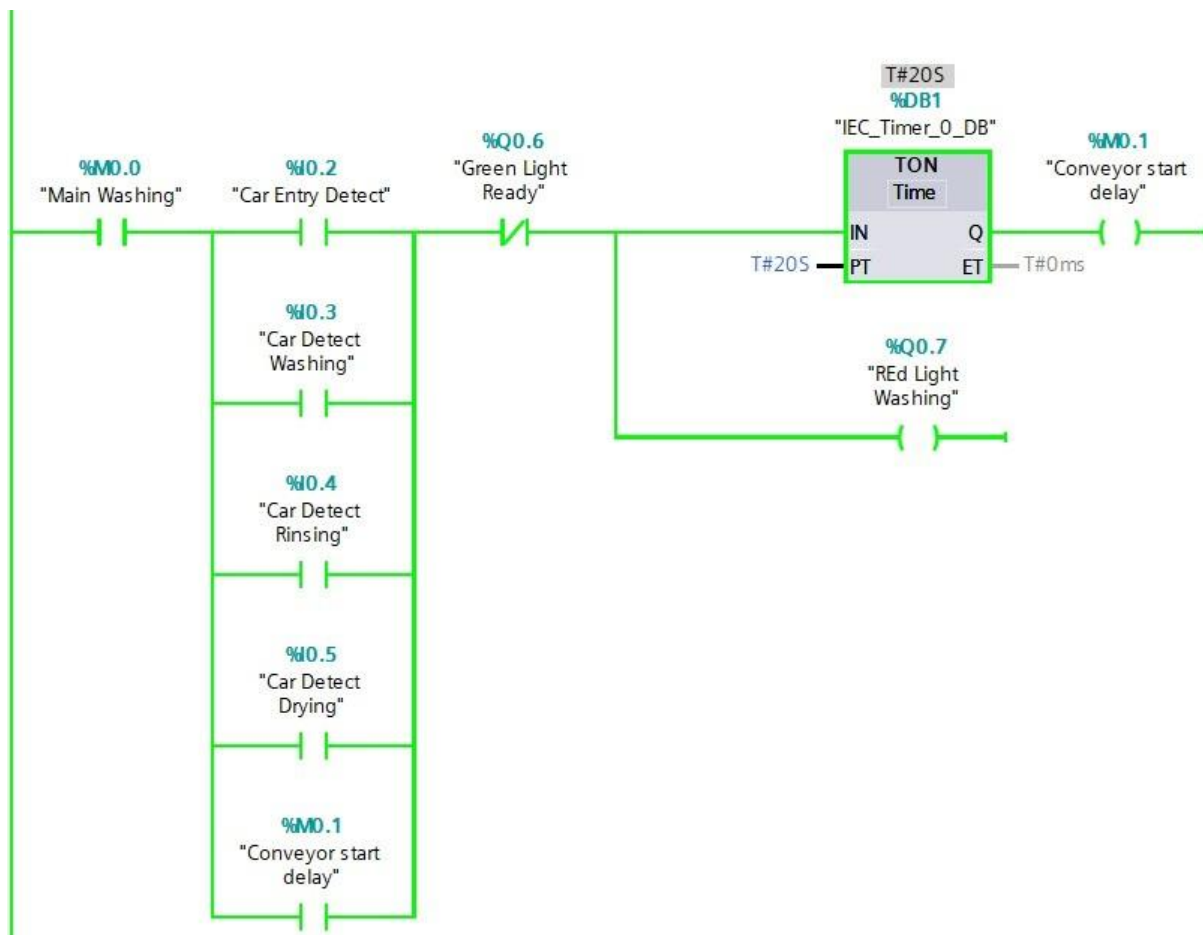
B) EXPERIMENT DETAILS (PROCEDURE)

Network 1: Washing ON and OFF



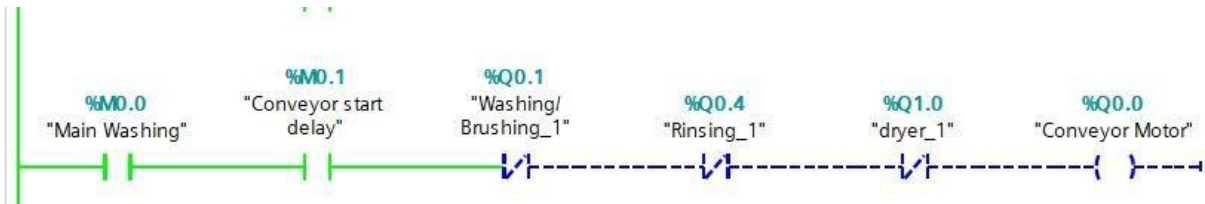
- **Inputs:** I0.0 (Start), I0.1 (Stop)
- **Memory Bit:** M0.0
- **Logic:** The washing process starts and stops based on the inputs.

Network 2: Conveyor Start Delay



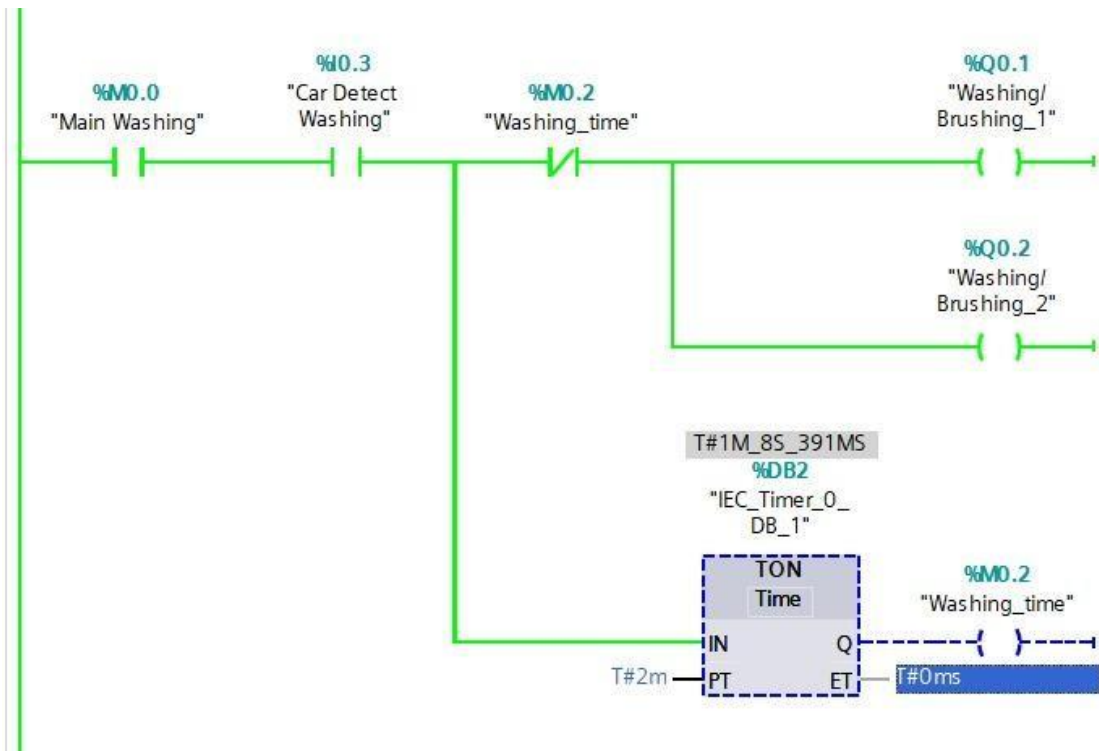
- **Input:** I0.2 (Car Entry Sensor)
- **Memory:** M0.1(Conveyor start)
- **Output:** Q0.7 (Washing in Progress Light)
- **Logic:** Detection of the car sets an on-delay timer for the conveyor motor and activates the red light, allowing the driver to exit the car before washing begins.

Network 3: Conveyor Motor



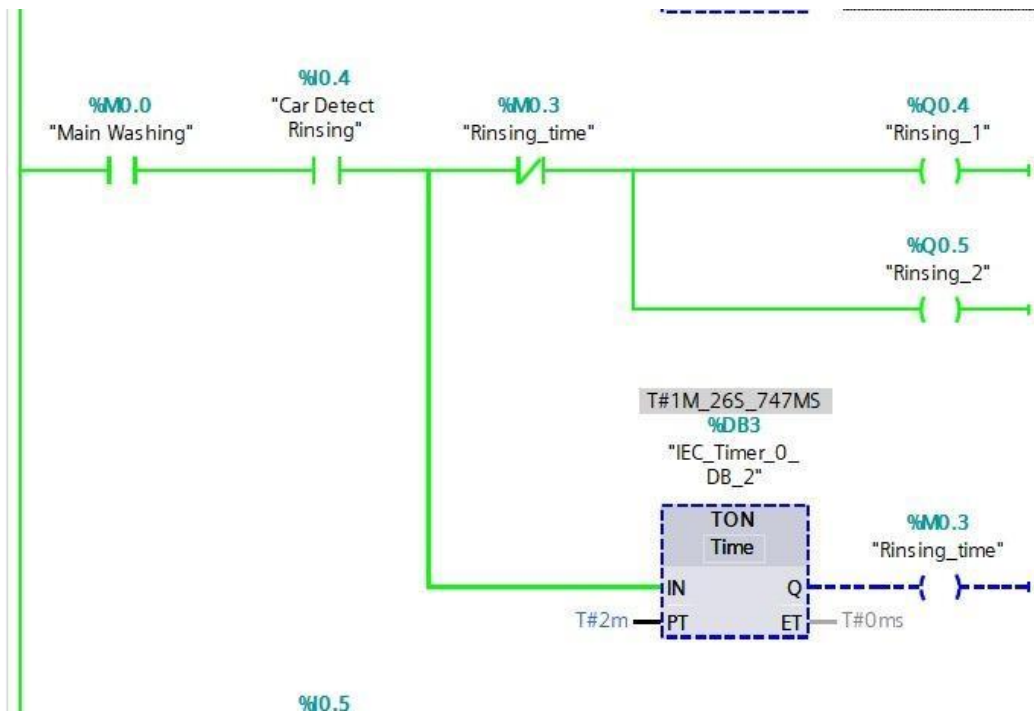
- **Input:** Timer
- **Output:** Q0.0 (Conveyor Motor)
- **Logic:** After the delay, the conveyor motor activates, moving the car to the washing stage.

Network 4: Car Washing and Brushing



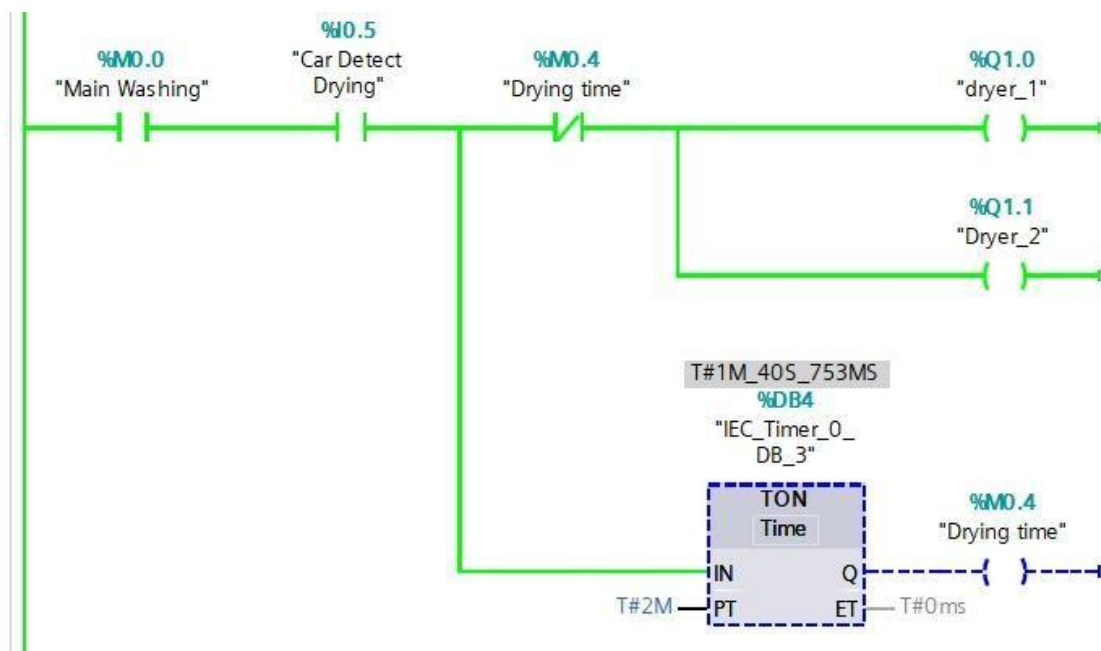
- **Input:** I0.3 (Car Washing Stage Sensor or Car Detect Washing)
- **Outputs:** Q0.1, Q0.2 (Washer/Brush)
- **Logic:** Washing and brushing start upon car detection and last for 2 minutes.

Network 5: Car Rinsing Logic



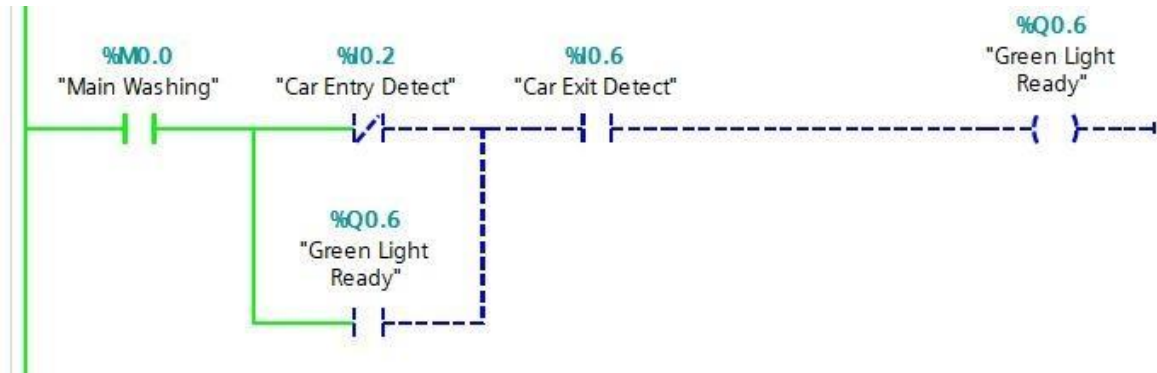
- **Input:** I0.4 (Car Rinsing Stage Sensor)
- **Output:** Q0.5, Q0.4 (Rinse)
- **Logic:** Rinsing starts upon car detection and lasts for 2 minutes.

Network 6: Car Drying Logic



- **Input:** I0.5 (Car Drying Stage Sensor)
- **Output:** Q1.0, Q1.1 (Dryer)
- **Logic:** Drying starts upon car detection and lasts for 2 minutes.

Network 7: Car Washing Complete

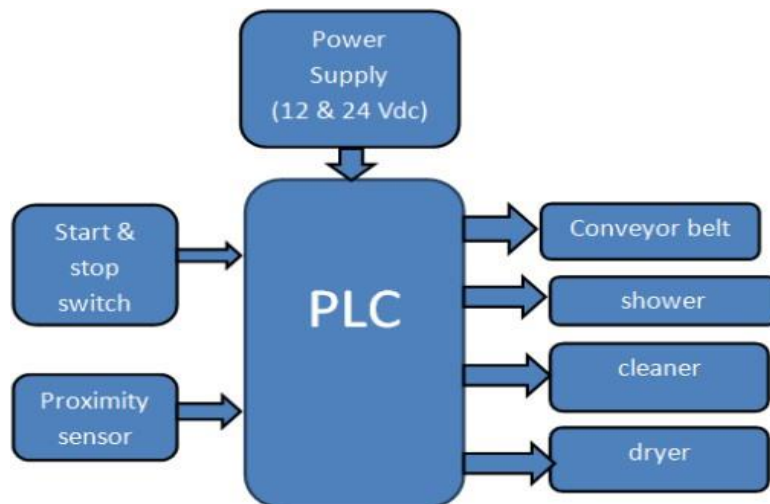
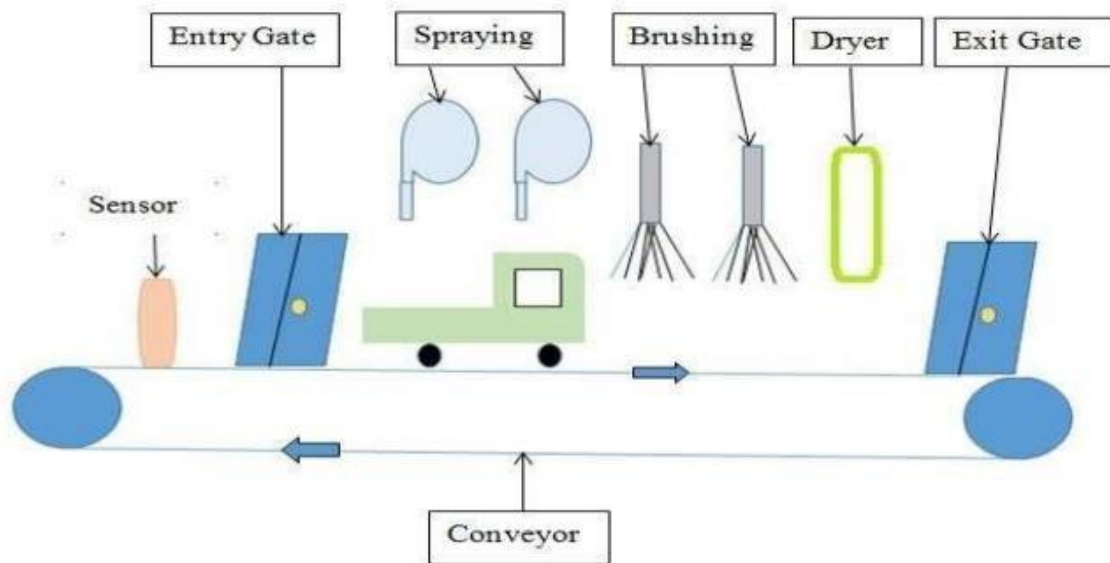


- **Input:** I0.6 (Car Exit Sensor)
- **Outputs:** Q0.6 (Washing Complete Light), Q0.0 (Conveyor Motor)
- **Logic:** The green light activates to signal completion, and the conveyor stops, allowing the car to exit.

Address	Description	IO types	Address	Description	IO types
I0.0	Start	Input	M0.0	Main Washing	Memory
I0.1	stop	Input	M0.1	Conveyor Start delay	Memory
I0.2	Car Entry Detect	Input	M0.2	Washing time	Memory
I0.3	Car Detect washing	Input	M0.3	Rinsing time	Memory
I0.4	Car Detect Rinsing	Input	M0.4	Drying time	Memory
I0.5	Car Detect Drying	Input	Q0.0	Conveyor start	Output
I0.6	Car Exit Detect	Input	Q0.1	Washing/Brashing_1	Output
Q0.2	Washing / Brashing_2	Output	Q0.4	Rinsing_1	Output
Q0.5	Rinsing_2	Output	Q0.6	Green Light ready	Output
Q1.0	Dryer_1	Output	Q1.1	Dryer_2	Output
Q0.7	Red Light Washing	Output			

RESULT

The implementation of the PLC-based automated carwash system demonstrated significant water savings and operational efficiency. The process control ensured precise timing and coordination of each stage, reducing human intervention and error. The system's ability to filter and reuse 75% of the water minimized the need for freshwater intake



CONCLUSION

The automated carwash system using PLC technology proved effective in conserving water and enhancing operational efficiency. The precise control offered by the PLC ensured a smooth and reliable washing process. The study demonstrated that automating carwash operations can significantly reduce water consumption and labor costs, making it a viable solution for addressing water scarcity and economic challenges in Kuwait and similar regions.

REFERENCES

- IEEE: Accession Number: 16866406 Design and Development of Automatic Carwash System Anam Abid , Tahir Hasan, Taimoor Baig, Abdullah Jadoon
- IEEE: Accession Number: 17860708 PLC based Automatic Car Washing System using Proximity Sensors Ragini Gaikwad, Sidhesh Mohite, Mayuri Kharat, Jayendra Thakur
- Johnson, C. D., Process Control Instrumentation Technology, Prentice Hall, 2006. Prof. Mhaske D.A., Bhavthankar R.G, Saindane A. R., Darade D.J.; “PLC Based Car Washing System”; International Journal Of Innovative Research In Electrical, Electronics, Instrumentation And Control Engineering Vol. 4, Issue 4, April 2016
- Al-Odwani, M., Ahmed, & Bou-Hamad, S. (2007). Carwash water reclamation in Kuwait. Desalination: EuroMed, Elsevier, 206(1-3), 17–28.